

How to solve for Big O questions

Question Type /Situation:

You will be given a function or an algorithm/ Pseudocode. And you will be asked to find upper bound of that function/algorithm/pseudocode along with the value of c and n_0

Q1. $f(n) = n^4 + 100 * n^2 + 50 * n + 200$. Find the upper bound of this function in terms of Big-O along with the values of c and n_0

Steps to solve this type of question.

Step 1: Guess a $g(n)$

- We need to find out most contributing term of $f(n)$. In the above question, it is n^4 .
- Consider any function as $g(n)$, whose growth rate is equal to or more than highest contributing factor of $f(n)$.
- say $g(n) = n^4$

Step 2: Guess a value of c

- say $c = 5$

Step 3: Draw a table for different values of n , $f(n)$, and $c * g(n)$ like below:

S.No	n	$f(n) = n^4 + 100n^2 + 50 * n + 200$	$c * g(n) = 5 * n^4$	
1	0	200	0	
2	1	351	5	
3	2	716	80	
4	3	1331	405	
5	4	2256	1280	
6	5	3575	3125	
7	6	5396	6480	Tipping point
8	7	7851	12005	
9	8	11096	20480	
10	9	15311	32805	
11	10	20700	50000	

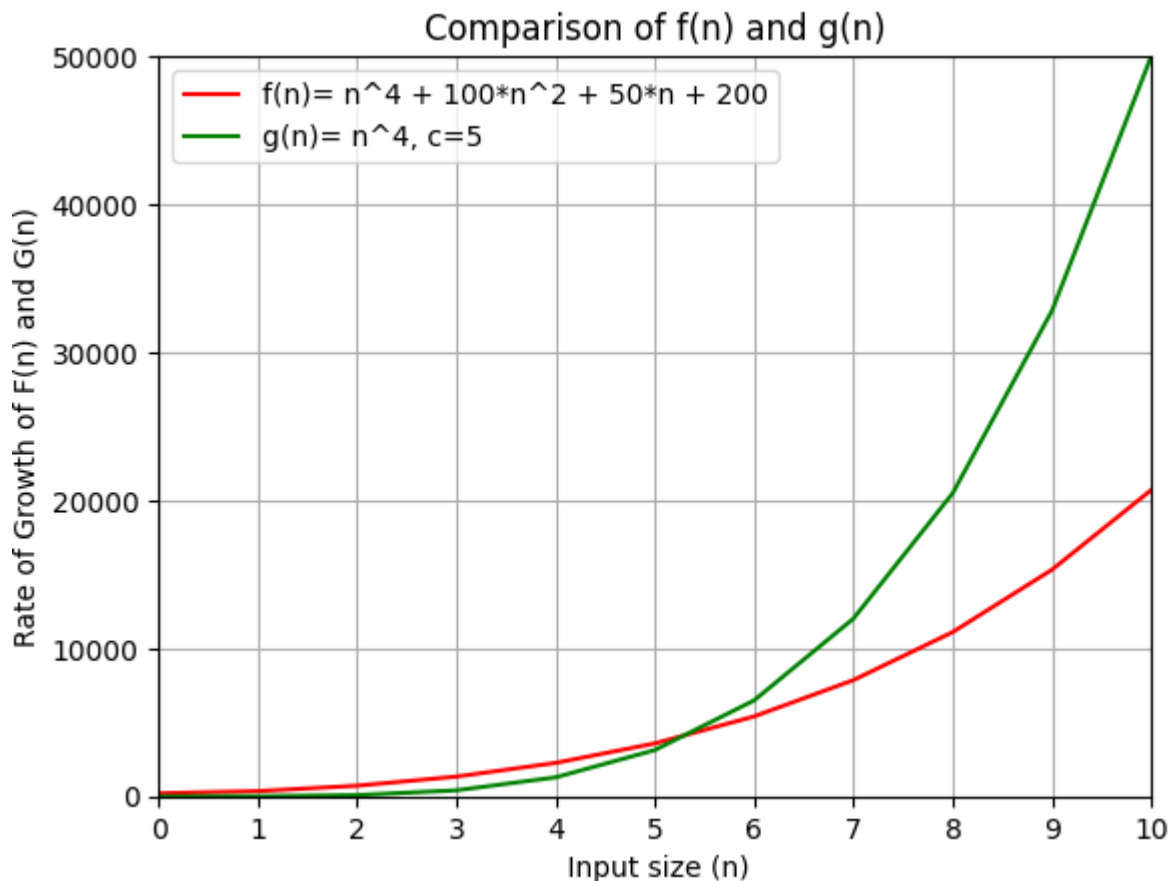
- We can now choose any value of n_0 that is greater than or equal to 6 (six). (say $n_0 = 9$)

Now $c * g(n) \geq f(n) \quad \forall n > n_0$

Thus, we can write, $f(n) = O(g(n))$

where $g(n) = n^4$ with $c = 5$, and $n_0 = 9$

Step 4: Plot the graph



Note:

- c and n_0 are not unique. Multiple values of c , and n_0 is possible for same value of $f(n)$ and $g(n)$.
- c and n_0 are +ve real constant.

if you cannot find appropriate value of c and n_0 for a given $g(n)$. Then modify your **guess of $g(n)$** by considering a function with higher growth rate.

Q2. $f(n) = 2 * n^3 - 2 * n^2$. Find the upper bound of this function in terms of Big-O along with the values of c and n_0 .

Solution:

Step 1: Let us guess $g(n) = n^3$

Step 2: Let us guess $c = 10$

Step 3: Draw a table for different values of n , $f(n)$, and $C * g(n)$ like below:

Q3. $f(n) = 2 * n^3 - 2 * n^2$. Find the upper bound of this function in terms of Big-O along with the values of c and n_0